

Probability PhD Exam, September 2009

1. For each fixed n , $X_n^1, X_n^2, \dots, X_n^n$ are independent random variables and have the same Binomial distribution, that is, $\mathbb{P}(X_n^i = 1) = p_n$ and $\mathbb{P}(X_n^i = 0) = 1 - p_n$ for $i = 1, 2, \dots, n$. Let $S_n = X_n^1 + X_n^2 + \dots + X_n^n$. Given the condition that np_n converges to a positive constant β as n goes to ∞ , find the limit distribution of S_n and prove it.

2. Let X and Y be two independent random variables with $\mathbb{E}[Y] = 0$. Show that for $p \geq 1$, we have

$$\mathbb{E}|X + Y|^p \geq \mathbb{E}|X|^p.$$

3. Let X be a positive random variable with a probability density function $f(x)$ that is continuous and $U_n = nX - [nX]$, where $[a]$ means the largest integer which is less than a . Find the limit distribution of U_n as n goes to ∞ .

4. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with probability density function $f(x)$, and $M = \max(X_1, X_2, \dots, X_n)$. Find the conditional expectation $\mathbb{E}[X_1 | M]$.

5. For a continuous function $f(x)$ on $[0, 1]$, define

$$P_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Show that $P_n(x)$ converges to $f(x)$ on $[0, 1]$ as n goes to infinity.

6. Let S_n be a one-dimensional simple random walk, that is, $S_n = X_1 + \dots + X_n$, where $X_1, \dots, X_n$ are i.i.d. random variables with $\mathbb{P}(X_1 = +1) = \mathbb{P}(X_1 = -1) = 1/2$, and $S_0 = 0$. and let

$$R_n = 1 + \max_{0 \leq k \leq n} S_k - \min_{0 \leq k \leq n} S_k$$

be the number of points visited by time n . Prove that $R_n/\sqrt{n}$ converges weakly to some limit (in distribution).

7. Let B_t be standard Brownian motion with $B_0 = 0$ and $\tau = \inf\{t : B_t = a + bt\}$, where $a > 0$ and b are two constants. Prove that

a).

$$\mathbb{E}_0[e^{-\lambda\tau}] = \exp\{-ab - a\sqrt{b^2 + 2\lambda}\}$$

b).

$$\mathbb{P}_0(\tau < \infty) = \exp\{-2ab\}.$$

8. Let $\{X_n, n = 0, 1, 2, 3, \dots\}$ be a Markov chain on a countable irreducible state space S . Let ϕ be a nonnegative function with

$$\mathbb{E}[\phi(X_1) | X_0 = x] \leq \phi(x)$$

for all but a finite number of x . Prove that if $\{x | \phi(x) \leq M\}$ is finite for any $M < \infty$, then the Markov chain $\{X_n, n = 0, 1, 2, 3, \dots\}$ is recurrent.